Data-Driven Analysis of the Indian Mutual Fund Industry: A Risk–Return and Investor Behaviour Study

Hariharan S
Department of Computer Science and Engineering
IIITDM Kurnool, Kallakurichi, Tamil Nadu, India
Bluestock Fintech -- Mutual Fund Analytics Capstone

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Abstract

This study presents an end-to-end analytical pipeline for the Indian mutual fund industry, covering 40 schemes across 10 fund houses, 46,000 net asset value (NAV) observations, 32,778 investor transactions, and 322 portfolio holdings spanning January 2022 to December 2025. We construct a star-schema SQLite data warehouse from ten heterogeneous source datasets and perform exploratory data analysis, risk-adjusted performance evaluation, and advanced risk analytics. Key contributions include: (i) a composite Fund Scorecard combining 3-year CAGR, Sharpe ratio, Jensen's alpha, expense ratio, and maximum drawdown into a single 0–100 ranking; (ii) Historical Value-at-Risk (VaR) and Conditional VaR (CVaR) estimates for all 40 schemes; (iii) a 90-day rolling Sharpe ratio analysis capturing the 2023 bull run and 2024 correction; (iv) investor cohort and SIP-continuity analysis identifying retention risk; and (v) a Herfindahl–Hirschman Index (HHI) based portfolio concentration study and a risk-appetite-driven fund recommender. Findings show that mid-cap equity funds dominate the top of the performance scorecard, small-cap funds carry the highest tail risk (VaR$_{95} \approx -2.4\%$ daily), and monthly SIP inflows grew from Rs. 11,517 crore (Jan 2022) to an all-time high of Rs. 31,002 crore (Dec 2025), alongside folio growth from 13.26 crore to 26.12 crore accounts.

1 Introduction

The Indian mutual fund industry has experienced sustained growth in assets under management (AUM), Systematic Investment Plan (SIP) participation, and retail folio counts over the period 2022–2025. For an asset manager such as Bluestock Fintech, the ability to systematically ingest heterogeneous data sources – fund master data, daily NAV history, transaction logs, portfolio holdings, and benchmark indices – and convert them into actionable risk and performance metrics is central to product design, investor communication, and regulatory reporting.

This paper documents a complete analytics pipeline built around ten raw datasets (Table 1), covering fund characteristics, daily NAV series, AUM by fund house, SIP and category inflows, industry folio counts, scheme-level performance indicators, investor-level transactions, portfolio holdings, and benchmark index levels. The pipeline is organised into five stages: data ingestion and quality validation, cleaning and warehousing, exploratory data analysis (EDA), performance analytics, and advanced risk analytics.

Table 1: Source datasets used in this study

Dataset	Rows	Granularity
Fund master	40	Per scheme
NAV history	46,000	Scheme × day
AUM by fund house	90	AMC × quarter
Monthly SIP inflows	48	Month
Category inflows	144	Category × month
Industry folio count	21	Month
Scheme performance	40	Per scheme
Investor transactions	32,778	Per transaction
Portfolio holdings	322	Scheme × stock
Benchmark indices	8,050	Index × day

2 Data Engineering and Quality Validation

2.1 Cleaning Procedures

Each dataset was cleaned independently before being loaded into a SQLite star-schema database (`bluestock_mf.db`). The principal cleaning steps were:

- **NAV history**: dates parsed to `datetime`, records sorted by AMFI code and date, duplicates removed, non-positive NAV values discarded, and the series reindexed onto a continuous daily calendar with forward-fill applied for weekends and holidays. This expanded the dataset from 46,000 to 64,320 rows.
- **Investor transactions**: transaction types standardised to $\{$*Sip*, *Lumpsum*, *Redemption*$\}$, amounts constrained to strictly positive values, and KYC status validated against the enumeration $\{$*Verified*, *Pending*, *Rejected*$\}$.
- **Scheme performance**: return and ratio columns coerced to numeric types, and expense ratios constrained to the plausible range 0.1%–2.5%.
- **Benchmark indices**: sorted by index name and date, duplicates removed, and daily returns derived from closing

values.

2.2 AMFI Code Validation

A consistency check confirmed that all 40 AMFI scheme codes present in the fund master dataset also appear in the NAV history dataset, and vice versa, indicating no orphaned or missing scheme references (Table 2).

Table 2: AMFI code cross-validation results

Check	Result
Fund master codes	40
NAV history codes	40
Codes missing from NAV history	0
Codes in NAV not in fund master	0
Validation status	**PASS**

2.3 Star Schema

The cleaned data was loaded into a SQLite database with one dimension table for funds (`dim_fund`), a generated date dimension (`dim_date`, 1,461 rows spanning 2022–2025), and six fact tables: `fact_nav`, `fact_transactions`, `fact_performance`, `fact_aum`, `fact_portfolio`, and `fact_benchmarks`. Ten analytical SQL queries were written against this schema, covering top funds by AUM, SIP year-over-year growth, state-wise transaction totals, low-cost funds (expense ratio $< 1\%$), and category-wise Sharpe ratio leaders.

3 Exploratory Data Analysis

The fund universe spans 10 fund houses and two broad categories (Equity and Debt) across 12 sub-categories, including Large Cap, Mid Cap, Small Cap, Flexi Cap, ELSS, Index/ETF, Gilt, Liquid, and Short Duration funds.

3.1 NAV Trends and Market Regimes

Figure 1 shows the NAV trajectories of five large-cap schemes from 2022 to 2026. The 2023 bull run is visible as a sustained upward trend, while the 2024 correction window shows a temporary flattening/decline before recovery into 2025.

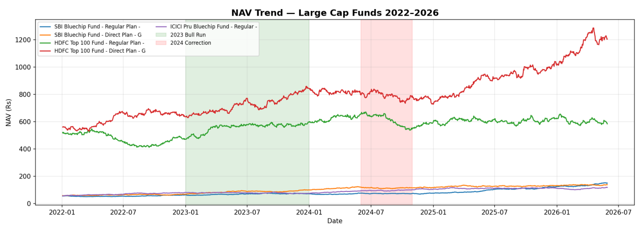

Figure 1: NAV trends for five large-cap schemes (2022–2026), with the 2023 bull run and 2024 correction periods highlighted.

3.2 AUM Concentration by Fund House

Figure 2 presents AUM by fund house across 2022–2025. Aggregating scheme-level AUM from the performance dataset, the total across all 40 schemes is approximately Rs. 10.44 lakh crore, with SBI Mutual Fund, ICICI Prudential, and HDFC Mutual Fund among the largest contributors.

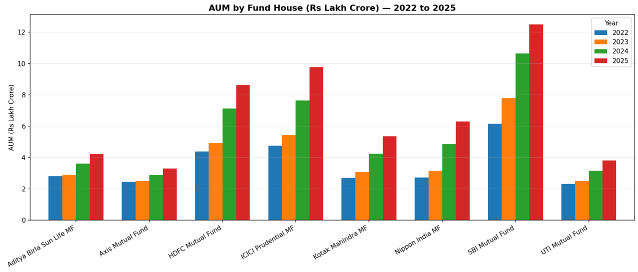

Figure 2: AUM by fund house (Rs. lakh crore), 2022–2025.

3.3 SIP Inflows and Folio Growth

Monthly SIP inflows grew from Rs. 11,517 crore in January 2022 to an all-time high of Rs. 31,002 crore in December 2025 (Figure 3), an increase of roughly 169% over the four-year window. Over the same period, industry folio counts nearly doubled, from 13.26 crore to 26.12 crore (Figure 4).

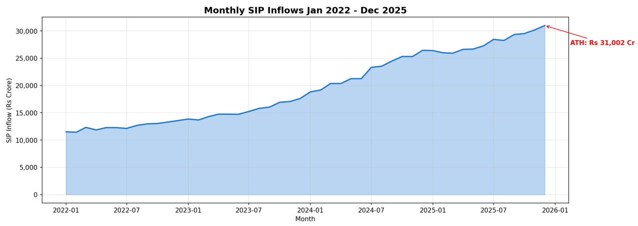

Figure 3: Monthly SIP inflows (Rs. crore), January 2022 – December 2025, with the all-time-high month annotated.

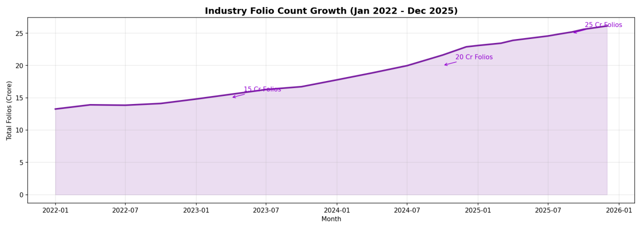

Figure 4: Industry folio count growth (crore accounts), Jan 2022 – Dec 2025.

3.4 Category Inflow Patterns

The category-level inflow heatmap (Figure 5) shows inflow intensity varying substantially across equity and debt categories over time, with equity-oriented categories generally showing stronger and more sustained positive net inflows than debt categories during the 2023–2025 window.

3.5 Sector Allocation and Return Correlation

Aggregating portfolio holdings across all equity schemes (Figure 6), Financials and Information Technology represent the two largest sector exposures, consistent with the composition of major Indian equity benchmarks. The pairwise return correlation matrix for 12 representative funds (Figure 7) shows correlations in the 0.7–0.95 range among large-cap and flexi-cap funds, indicating limited diversification benefit when combining funds from similar sub-categories.

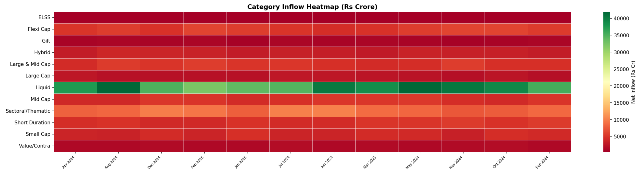

Figure 5: Net inflow by category and month (Rs. crore); green indicates positive net inflow, red indicates outflow.

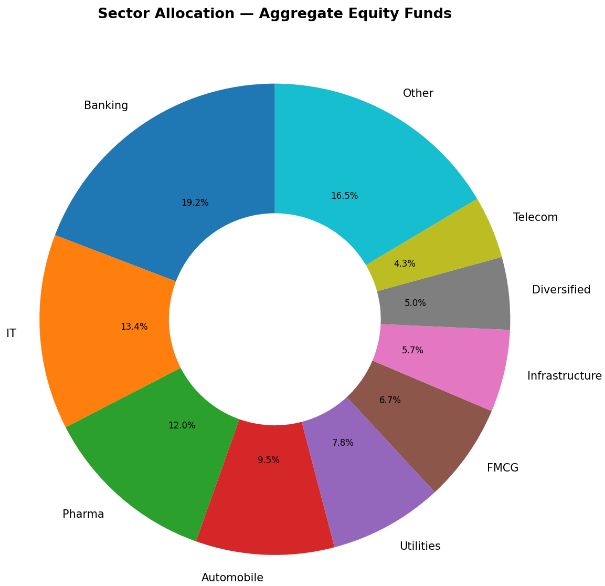

Figure 6: Aggregate sector allocation across equity scheme holdings.

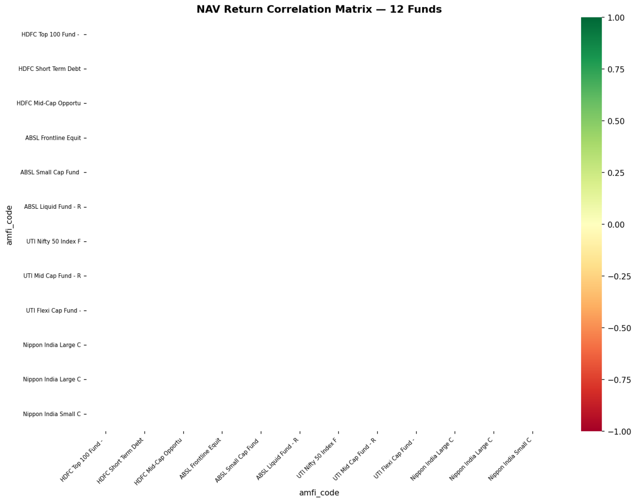

Figure 7: Pairwise correlation matrix of daily returns for 12 representative schemes.

4 Performance Analytics

4.1 Methodology

For each scheme i, the daily return is defined as

$$r_{i,t} = \frac{\mathrm{NAV}_{i,t}}{\mathrm{NAV}_{i,t-1}} - 1. \tag{1}$$

CAGR over n years is computed as

$$\mathrm{CAGR}_i = \left(\frac{\mathrm{NAV}_{i,\mathrm{end}}}{\mathrm{NAV}_{i,\mathrm{start}}} \right)^{1/n} - 1. \tag{2}$$

The Sharpe and Sortino ratios use a risk-free rate $R_f = 6.5\%$ (a proxy for the RBI repo rate):

$$\mathrm{Sharpe}_i = \frac{\bar{r}_i - R_f^d}{\sigma_i}\sqrt{252}, \qquad \mathrm{Sortino}_i = \frac{\bar{r}_i - R_f^d}{\sigma_i^-}\sqrt{252}, \tag{3}$$

where σ_i^- is the standard deviation of negative excess-return days only. Jensen's alpha and beta are estimated via ordinary least squares regression of fund returns on NIFTY 100 returns, $r_{i,t} = \alpha_i + \beta_i r_{m,t} + \varepsilon_t$, with $\alpha_i^{\mathrm{ann}} = \alpha_i \times 252$. Maximum drawdown is $\mathrm{MDD}_i = \min_t \left(\mathrm{NAV}_{i,t} / \max_{s \le t} \mathrm{NAV}_{i,s} - 1 \right)$.

4.2 Composite Fund Scorecard

A composite score on a 0–100 scale was constructed for each scheme as a weighted combination of percentile ranks:

$$\begin{aligned}\mathrm{Score}_i = {} & 0.30 \cdot \mathrm{rank}(\mathrm{CAGR}_{3y,i}) + 0.25 \cdot \mathrm{rank}(\mathrm{Sharpe}_i) \\ & + 0.20 \cdot \mathrm{rank}(\alpha_i^{\mathrm{ann}}) + 0.15 \cdot \mathrm{rank}^{-1}(\mathrm{ExpRatio}_i) \\ & + 0.10 \cdot \mathrm{rank}^{-1}(\mathrm{MDD}_i),\end{aligned} \tag{4}$$

where rank^{-1} denotes an inverted percentile rank (lower expense ratio and shallower drawdown score higher). Table 3 reports the top five schemes by composite score.

Table 3: Top 5 schemes by composite Fund Scorecard

Scheme	CAGR3y	Sharpe	Score
ICICI Pru Midcap	31.78%	0.883	84.50
Axis Midcap	35.11%	0.731	81.38
HDFC Mid-Cap Opp.	32.44%	0.808	79.88
Mirae Asset Large Cap	34.00%	1.068	79.38
Kotak Flexicap	29.58%	0.966	77.62

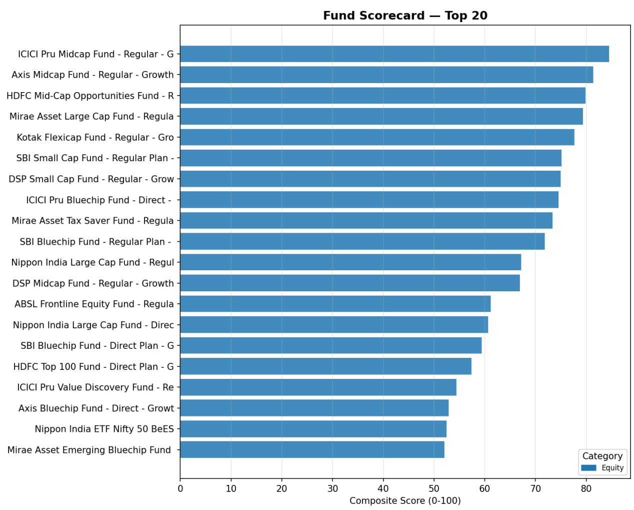

Figure 8: Top 20 schemes ranked by composite Fund Scorecard, coloured by category.

Mid-cap equity funds dominate the top of the scorecard, reflecting strong 3-year CAGR figures (29–35%) combined with competitive Sharpe ratios. The schemes with the highest annualised alpha against NIFTY 100 were SBI Small Cap (+30.34%), DSP Small Cap (+30.06%), and ICICI Prudential Midcap (+29.26%), each exhibiting betas close to zero, suggesting return drivers largely independent of the large-cap benchmark over the sample window.

4.3 Drawdown Risk

The deepest maximum drawdowns were observed in small-cap schemes: SBI Small Cap Direct Plan (-52.6%), Axis Small Cap (-51.7%), and ABSL Small Cap (-35.5%), all with trough dates falling in late 2025 / mid-2026, consistent with elevated small-cap volatility relative to large-cap and flexi-cap peers.

4.4 Benchmark Comparison

Figure 9 compares the top five scorecard funds against NIFTY 50 and NIFTY 100 on a rebased (=100) cumulative basis. All five funds outperformed both benchmarks over the sample window, with the tracking error for the top performers in the range of moderate single digits, reflecting active deviation from benchmark composition consistent with mid-cap and flexi-cap mandates.

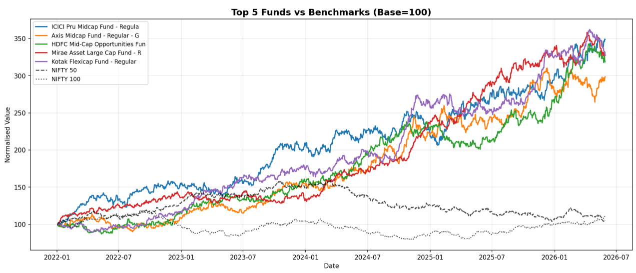

Figure 9: Top 5 scorecard funds vs. NIFTY 50 and NIFTY 100 (rebased to 100).

5 Advanced Risk and Behavioural Analytics

5.1 Value-at-Risk and Conditional VaR

Historical 95% VaR was computed as the 5th percentile of the daily return distribution, with CVaR as the mean of returns falling below this threshold. Figure 10 shows the 15 riskiest schemes by VaR_{95}. The highest-risk schemes are exclusively small-cap funds: ABSL Small Cap ($\text{VaR}_{95} = -2.39\%$, $\text{CVaR}_{95} = -3.03\%$), Axis Small Cap ($-2.33\%$ / -2.97%), and SBI Small Cap Direct (-2.32% / -3.02%), each with annualised volatility around 21%.

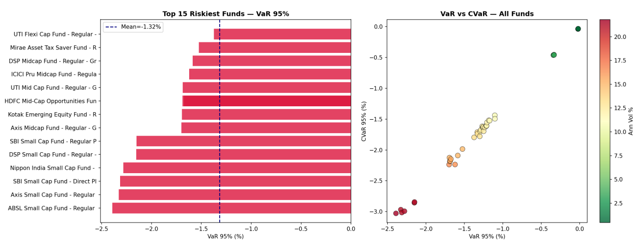

Figure 10: Left: Top 15 schemes by VaR_{95}. Right: VaR vs. CVaR scatter coloured by annualised volatility.

5.2 Rolling 90-Day Sharpe Ratio

The rolling 90-day Sharpe ratio for the five most data-complete schemes (Figure 11) confirms the regime shifts visible in the NAV trend chart: Sharpe ratios compress toward zero or turn negative during the 2024 correction window before recovering through 2025, illustrating time-varying risk-adjusted performance that a single full-sample Sharpe ratio would obscure.

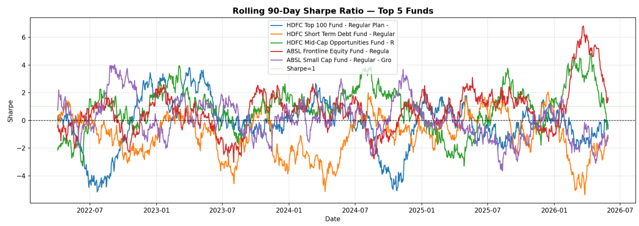

Figure 11: Rolling 90-day Sharpe ratio for the five schemes with the longest NAV history.

5.3 Investor Cohort Analysis

Investors were grouped by the calendar year of their first recorded transaction. The resulting cohorts (Table 4) show that the 2024 cohort accounts for the overwhelming majority of investors (4,803) and transaction volume (32,499 transactions, totalling approximately Rs. 349 crore invested), with an average SIP ticket size of roughly Rs. 1,074. The smaller 2025 cohort shows a marginally higher average SIP amount (Rs. 1,092), and both cohorts' top fund preference includes passive (Nifty 50 Index) and small-cap active strategies.

Table 4: Investor cohort summary by first-transaction year

Cohort	Investors	Txns	Avg. SIP (Rs.)
2024	4,803	32,499	1,074.23
2025	197	279	1,091.59

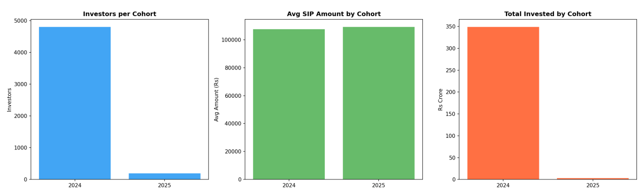

Figure 12: Investor count, average SIP amount, and total invested by first-transaction-year cohort.

5.4 SIP Continuity

Among investors with six or more SIP transactions, the average gap between consecutive SIPs was computed; investors with an average gap exceeding 35 days were flagged as *at-risk* of SIP discontinuation. Of 1,362 qualifying investors, a substantial majority were flagged as at-risk under this threshold, indicating that irregular SIP cadence is widespread in the sample and represents a meaningful retention-management opportunity for fund houses and distributors.

5.5 Sector Concentration (HHI)

Portfolio concentration was measured via the Herfindahl–Hirschman Index, $\text{HHI}_i = \sum_k w_{i,k}^2$, over sector weights

$w_{i,k}$ for each equity scheme i. Figure 13 shows that the Axis Bluechip Fund and ABSL Small Cap Fund exhibit the highest concentration (HHI $\approx$ 2,000–2,065, *Moderate* band), while Nippon India Small Cap and both SBI Bluechip variants show the lowest concentration (HHI $\approx$ 1,073–1,084, *Low* band). No scheme in the sample crossed the 2,500 *High* concentration threshold, suggesting reasonably diversified sector exposure across the fund universe relative to conventional concentration benchmarks.

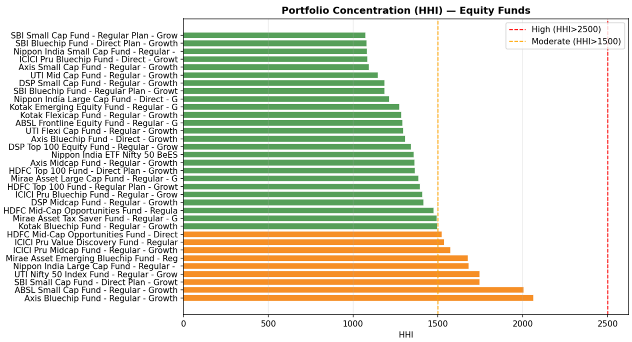

Figure 13: Herfindahl–Hirschman Index by equity scheme, with Moderate (HHI>1500) and High (HHI>2500) thresholds annotated.

5.6 Risk-Appetite-Based Fund Recommender

A rule-based recommender filters the Fund Scorecard by an investor's stated risk appetite – Low, Moderate, or High – mapped onto the dataset's `risk_grade` field (e.g. Moderate maps to {*Moderate*, *Moderately High*}), and returns the top three schemes by Sharpe ratio within that band. For a *Moderate* risk appetite, the recommender returns Mirae Asset Large Cap Fund (Sharpe $= 1.068$), Kotak Flexicap Fund (Sharpe $= 0.966$), and SBI Bluechip Fund (Sharpe $= 0.861$), illustrating how the composite scorecard can be operationalised for investor-facing recommendations.

6 Risk–Return Trade-off Summary

Figure 14 consolidates the risk and return dimensions across all 40 schemes, plotting annualised standard deviation against 3-year CAGR with bubble size proportional to AUM and colour representing the Sharpe ratio. The plot visually corroborates the category-level findings: small-cap schemes occupy the high-risk / high-return region with generally lower Sharpe ratios, while large-cap and flexi-cap schemes cluster in a more favourable lower-risk region with comparatively higher Sharpe ratios, consistent with their dominance of the composite scorecard.

7 Discussion

Three themes emerge from this analysis. First, the performance–risk trade-off is strongly category-driven: mid-cap and flexi-cap funds achieve the best risk-adjusted returns in the sample, while small-cap funds, despite occasionally posting the highest raw alpha, carry disproportionately higher tail risk (VaR_{95} around -2.3% daily and drawdowns exceeding -50%). Second, industry growth metrics – SIP inflows nearly tripling and folio counts doubling over four years – indicate sustained retail participation, but the SIP continuity analysis suggests that a large share of active SIP

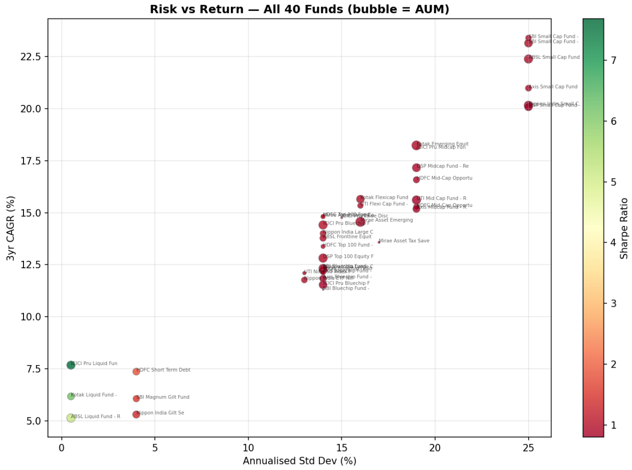

Figure 14: Risk (annualised std. dev.) vs. 3-year CAGR across all 40 schemes; bubble size = AUM, colour = Sharpe ratio.

investors exhibit irregular contribution patterns, which has direct implications for AUM stability projections. Third, sector concentration as measured by HHI remains within the *Low* to *Moderate* bands across the sample, suggesting that, at least at the portfolio level, equity schemes in this dataset are reasonably diversified relative to common concentration thresholds.

8 Limitations and Future Work

The analysis relies on a fixed 4-year window (2022–2025) and a 40-scheme universe across 10 fund houses, which may not be representative of the full industry (1,900+ schemes). The fund recommender uses a simple rule-based filter rather than a personalised, multi-factor optimisation; future work could incorporate Markowitz mean-variance optimisation for portfolio construction (Efficient Frontier analysis), Monte Carlo simulation of NAV paths for forward-looking projections, and a richer investor-segmentation model combining demographic, behavioural, and risk-tolerance signals. Automating the live NAV ingestion (via the mfapi.in API) on a scheduled basis would also enable the dashboard and analytics layers to operate on near-real-time data.

9 Conclusion

This study delivers a complete, reproducible analytics pipeline – from raw CSV ingestion through a SQLite star schema to composite fund scoring and advanced risk analytics – applied to a realistic Indian mutual fund dataset. The composite Fund Scorecard, VaR/CVaR risk profiling, rolling Sharpe analysis, investor cohort study, SIP continuity flagging, and HHI concentration analysis together provide a multi-dimensional view of fund performance, risk, and investor behaviour suitable for both internal analytics teams and investor-facing recommendation systems.

Acknowledgements

This work was completed as part of the Bluestock Fintech Mutual Fund Analytics capstone project. All datasets, cleaned tables, notebooks, SQL scripts, and generated figures are available in the accompanying project repository.